§3 Prob:- Two equal elastic spheres each of mass m move in the same direction with equal velocities u and collide directly. Show that they interchange their velocities.

Ans. Let v_1 and v_2 be the velocities of the spheres after impact. By Newton's law on impact,

$$v_2 - v_1 = e(u-u) = 0$$

$$\Rightarrow v_2 = v_1$$

Also, by the principle of conservation of linear momentum,

$$m v_1 + m v_2 = mu + mu$$

$$\Rightarrow v_1 = v_2 = u$$

∴ So, the velocities are interchanged.

§ Prob: 4:- Two elastic spheres impinges directly with equal and opposite velocities. Find the ratio of their masses so that one of these may come to rest after impact, the ~~co~~ co-efficient of resolution being e.

Ans. Let m_1 & m_2 be the masses of the spheres and u be their velocities (in opposite directions).

One of the spheres comes to rest after impact and let the velocity of the ~~third~~ sphere be v after impact.

Then by Newton's law on impact

$$0 - v = e(u+u) = 2ue$$

$$\text{i.e. } v = -2eu$$

By the principle of conservation of linear momentum.

$$m_1 v = m_1 u - m_2 u$$

$$\text{or, } (1+2e) m_1 u = m_2 u$$

$$\text{or } m_1 : m_2 = 1 : (1+2e)$$

§ Prob 5:- A spheres impinges directly on an equal sphere at rest. If the co-efficient of restitution be e, then show that their velocities after impact are in the ratio $(1-e):(1+e)$.

Ans Let u be the velocity of the first sphere before impact and m be masses of both the spheres. Also, let v_1 and v_2 be their velocities after impact.

By Newton. experimental law

$$v_2 - v_1 = e(u-0) = eu \quad \text{——} (i)$$

Also, by the principle of conservation of linear momentum.

$$mu = mv_2 + mv_1$$

$$\Rightarrow u = v_2 + v_1 \quad \text{——} (ii)$$

. From (i) & (ii) $\quad \frac{v_2 - v_1}{v_2 + v_1} = \frac{eu}{u}$

$$\Rightarrow \frac{2v_2}{2v_1} = \frac{e+1}{1-e}$$

$$\Rightarrow \frac{v_1}{v_2} = \frac{1-e}{1+e}.$$

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